

Diet and risk of ischemic heart disease in India.

BACKGROUND: Ischemic heart disease (IHD) is a leading cause of death in India. Dietary changes could reduce risk, but few studies have addressed the association between diet and IHD risk in India.

OBJECTIVE: The goal was to address the association between diet and IHD risk among Indians in New Delhi (northern India) and Bangalore (southern India).

DESIGN: We collected data from 350 cases of acute myocardial infarction and 700 controls matched on the basis of age, sex, and hospital as part of a hospital-based case-control study in 8 hospitals. Long-term dietary intake was assessed by using food-frequency questionnaires developed for New Delhi and Bangalore. We used conditional logistic regression to control for the matching factors and other predictors of risk.

RESULTS: We observed a significant and dose-dependent inverse association between vegetable intake and IHD risk. The inverse association was stronger for green leafy vegetables; in multivariate analysis, persons consuming a median of 3.5 servings/wk had a 67% lower relative risk (RR: 0.33; 95% CI: 0.17, 0.64; P for trend = 0.0001) than did those consuming 0.5 servings/wk. Controlling for other dietary covariates did not alter the association.

Cereal intake was also associated with a lower risk. Use of mustard oil, which is rich in alpha-linolenic acid, was associated with a lower risk than was use of sunflower oil [for use in cooking: RR: 0.49 (95% CI: 0.24, 0.99); for use in frying, RR: 0.29 (95% CI: 0.13, 0.64)].

CONCLUSION: Diets rich in vegetables and use of mustard oil could contribute to the lower risk of IHD among Indians.